

BIRTHDAY GIFT



BY PIYUSH SIR

FORMULA E-book

by Piyush Sir

ANALOGY
A : B :: C : ?

SERIES
2, 4, 8, 16, ?

CODING
DECODING



DIRECTION

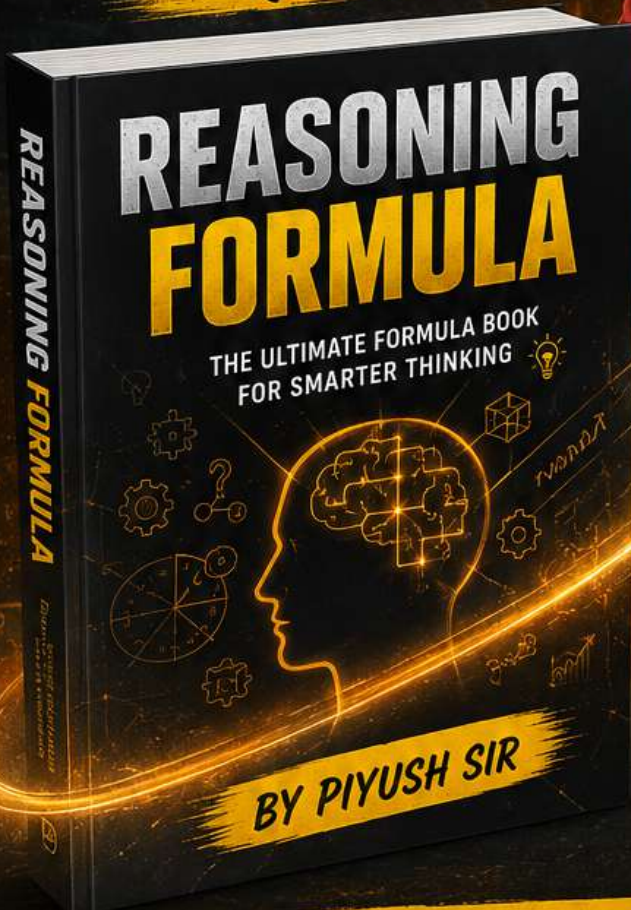
BLOOD
RELATION



ALPHABET
TEST



PUZZLES



- ✓ SHORTCUTS & TRICKS
- ✓ CHAPTER-WISE FORMULAS
- ✓ EASY TO LEARN
- ✓ SAVE TIME, SCORE MORE!

LEARN SMART, SOLVE FASTER,
ACHIEVE MORE!

Coding-Decoding

Short Notes

Coding is a method to represent something by some other symbol or sign. To easily solve the questions that come up in this chapter, we should remember the places of letters.

There are 3 types of questions asked in this chapter:

- (I) Letter Coding
- (II) Number Coding
- (III) Mixed Coding

I. Letter Coding

In these questions, the letters of a word are written in a coded language following a certain rule.

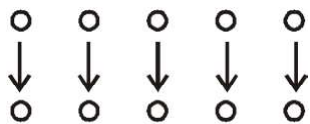
We have to recognize the pattern and answer it. Thus solving all the questions we have to find out the logical sequence.

There are 7 types of patterns in letter coding:

- (1) Forward pattern
- (2) Reverse pattern
- (3) Cross pattern
- (4) Position pattern
- (5) Opposite letter pattern
- (6) Common pattern
- (7) Illogical pattern

1. Forward Pattern:

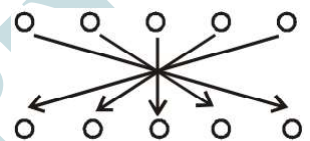
In this pattern, the first letter is changed into the first letter, the second letter into the second, the third letter into the third and in this way all the letters will be changed. A logical sequence has to be followed.



In this, any logical pattern can be formed. For e.g. It may be there is done +1 in all the letters or +2 or +3 or -1 or -2 or -3 or mixture of these all.

2. Reverse Pattern:

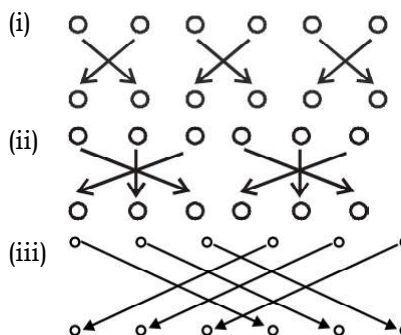
In this pattern, the first letter is changed into the last letter, the second letter is changed into the second last letter and in this way all the letters will be changed.



In this, any logical pattern can be formed. For e.g. It may be there is done +1 in all the letters or +2 or +3 or -1, -2, -3 or mixture of these all.

3. Cross Pattern:

In this pattern, the pairs are formed in cross form. In this, many types of arrangements can be made such as



4. Position Pattern:

No new letter will be made in this pattern. Neither we will add anything nor we will subtract anything. Only the places of letters will be changed. Hence there is no need to write the place value of letters.

5. Opposite letter Pattern:

In this pattern we convert the letters into their opposite letters. The two letters which are opposite to each other, have the sum of 27. Therefore by subtracting a letter's place value from 27, we can find out its opposite letter.

We can use the following trick to remember the opposite letters:

A → Z	
B → Y	(BYE)
C → X	(SIX)
D → W	(DEW)
E → V	(EVENING)
F → U	(FOUR)
G → T	(G.T. ROAD)
H → S	(High SCHOOL)
I → R	(Indian RAILWAY)
J → Q	(Jungle QUEEN)
K → P	(KANPUR)
L → O	(LOVE)
M → N	(MAN)

In this also, neither we will add anything nor we will subtract anything.

6. Common Pattern:

In this pattern, the letters which are common in the two words given, will also have the same code. Thus we don't apply any new logic in such questions. The letter which is written just below a letter will be the code of that letter.

7. Illogical Pattern:

When there is no logic in the question, then there will be an illogical pattern.

II. Number Coding

In this we use digits to represent a word in code language. In this also we perform coding by following a logical sequence.

There are mainly two types of patterns in it:

1. First Pattern:

In this pattern, we together form a code for the whole word. It can contain many types of patterns:

- In this pattern we add the values of all the letters of a word.
- In this, it is multiplied by a number after adding all the letters of the word. This number can be a variable or constant. The variable should also be logical and it can be number of letters or vowels or consonants in a word.
- In this, it is divided by a number after adding all the letters of the word. This number can be a variable or constant.
- In this, the coding is performed by adding the opposite values of all letters of a word.
- In this, it is multiplied by a number after adding the opposite values of all letters of a word. This number can be a variable or constant.
- In this, it is divided by a number after adding the opposite values of all letters of a word. This number can be a variable or constant.

2. Second Pattern:

In this pattern, while coding, the letters of the word will be used individually.

III. Mixed Coding

There are mainly two types of patterns in it:

1. Old Pattern:

The questions related to this pattern are asked in all competitive examinations.

2. New Pattern:

It has been asked only in banking exams.

Binary Coding

It comes under number coding. We use decimal number system which consists of 10 digits (0 to 9) but a computer understands the binary language which has only two digits (0,1), because the computers work on signals which are either on or off. When the signal is on, it means 1 and when the signal is off, it means 0.

The method of converting the decimal number system into binary:

- 2^7 2^6 2^5 2^4 2^3 2^2 2^1 2^0
.....128 64 32 16 8 4 2 1
- To convert a number into a binary code, write 1 below those powers of 2 which have been used and write 0 below those powers of 2 which have not been used.

For e.g. 21 will be converted into binary code in the following way:

$$\begin{array}{r} 2^7 \quad 2^6 \quad 2^5 \quad 2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0 \\ 128 \quad 64 \quad 32 \quad 16 \quad 8 \quad 4 \quad 2 \quad 1 \\ \hline 1 \quad 0 \quad 1 \quad 0 \quad 1 \end{array}$$

$$21 = 16 + 4 + 1$$

i.e. 2^4 , 2^2 and 2^0 have been used to write 21 into Binary. Therefore write 1 below them. While 2^3 and 2^1 have not been used, therefore write 0 below them.

$$(21)_{10} = (10101)_2$$

The method of converting binary into decimal number system:

- First write the given number.
- Starting from 2^0 below the given number, we will write the powers of 2 from right to left in increasing order.
- Those powers of 2 will be included above which 1 is written and those powers of 2 will not be included above which 0 is written. For e.g.

Convert $(11101)_2$, into decimal number system:

$$\begin{array}{r} 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ 2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0 \\ 16 \quad 8 \quad 4 \quad \quad 1 \\ 16 + 8 + 4 + 1 = 29 \end{array}$$

Convert $(111011)_2$ into decimal number system:

$$\begin{array}{r} 1 \quad 1 \quad 1 \quad 0 \quad 1 \quad 1 \\ 2^5 \quad 2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0 \\ 32 \quad 16 \quad 8 \quad \quad 2 \quad 1 \\ 32 + 16 + 8 + 2 + 1 = 59 \end{array}$$

Letter Series Short Notes

In the questions coming in these, usually a series of letters are given which follows a certain pattern. However, some letters are missing from such series. We have to find the missing letters by understanding the pattern. To solve such questions, the entire series is divided into some groups. These groups can be of any number of letters. Some possible groups can be as follows-

इनमें आने वाले प्रश्नों में सामान्यतः अक्षरों की एक श्रृंखला दी गई होती है जो कि किसी निश्चित पैटर्न का अनुसरण करती है। हालांकि ऐसी श्रृंखला में से कुछ अक्षर लुप्त होते हैं। हमें पैटर्न को समझकर लुप्त अक्षर ज्ञात करने होते हैं। ऐसे प्रश्नों को हल करने के लिए पूरी श्रृंखला को कुछ समूहों में बाँट लिया जाता है। ये समूह कितने भी अक्षरों के हो सकते हैं। कुछ संभावित समूह निम्न प्रकार से हो सकते हैं-

1. When all the groups formed are similar-

जब बनने वाले सभी समूह एक जैसे हो-

(i) Groups of two letters/दो-दो अक्षरों के समूह

ab/ab/ab/ab/.....

(ii) Groups of three letters each/तीन-तीन अक्षरों के समूह

abc/abc/abc/abc/.....

(iii) Groups of four letters/चार-चार अक्षरों के समूह

abcd/abcd/abcd/abcd/.....

Similarly, groups of letters 5, 6, 9 and 9 will be formed./इसी प्रकार 5, 6, अक्षरों के समूह बनेंगे।

2. When the alternative groups formed are similar-

जब बनने वाले वैकल्पिक समूह एक जैसे हों-

abc/xyz/abc/xyz/.....

3. When all the groups are different-

जब सभी समूह अलग-अलग हों-

Such groups will also be in some logical order.

ऐसे समूह भी किसी तर्कसंगत क्रम में होंगे।

abc/bca/cab/.....

4. When the number of letters in each group formed is different-

जब बनने वाले प्रत्येक समूह में अक्षरों की संख्या अलग-अलग हो-

ab/abc/abcd/abcde/.....

अक्षर श्रृंखला से संबंधित प्रश्नों को हल करने के कोई निश्चित नियम नहीं है लेकिन फिर भी हम कुछ ट्रिक्स/अवधारणाओं की सहायता से प्रश्नों को हल करने का प्रयास करेंगे।

There are no fixed rules to solve questions related to letter series but still we will try to solve the questions with the help of some tricks/concepts.

(A) First of all, we will find the total number of letters in the given letter series including spaces.

सबसे पहले दी गई अक्षर श्रृंखला में रिक्त स्थानों को शामिल करते हुए कुल अक्षरों की संख्या ज्ञात करेंगे।

(B) Now the entire series will be divided into groups based on the total number of letters.

अब कुल अक्षरों की संख्या के आधार पर पूरी श्रृंखला को समूहों में बाँटा जाएगा।

1. If the number of letters is '12' then we will make groups of '3' or '4' letters.

यदि अक्षरों की संख्या '12' है तो हम '3' या '4' अक्षरों के समूह बनाएंगे।

2. If the number of letters is '15' then we will make groups of '3' or '5' letters.

यदि अक्षरों की संख्या '15' है तो हम '3' या '5' अक्षरों के समूह बनाएंगे।

3. If the number of letters is '16' then we will make groups of 4 letters each.

यदि अक्षरों की संख्या '16' है तो हम 4-4 अक्षरों के समूह बनाएंगे।

4. If the number of letters is '18' then we will make groups of '3' or '6' letters.

यदि अक्षरों की संख्या '18' है तो हम '3' या '6' अक्षरों के समूह बनाएंगे।

5. If the number of letters is '20' then we will make groups of '4' or '5' letters.

यदि अक्षरों की संख्या '20' है तो हम '4' या '5' अक्षरों के समूह बनाएंगे।

6. If the number of letters is '28' then we will make groups of 7 letters.

यदि अक्षरों की संख्या '28' है तो हम 7 अक्षरों के समूह बनाएंगे।

7. If the number of letters is '32' then we will make groups of 8 letters.

यदि अक्षरों की संख्या '32' है तो हम 8 अक्षरों के समूह बनाएंगे।

(If the number of letters is 13,14,17,19, then such letter series will be solved in a different way.)

(यदि अक्षरों की संख्या 13/14/17/19 है तो ऐसी अक्षर श्रृंखलाओं को अलग तरीके से हल किया जाएगा।)

Keep in mind that the process of grouping based on the total number of letters is a concept, not a fixed rule.

यह ध्यान रखें कि कुल अक्षरों की संख्या के आधार पर समूह बनाने की प्रक्रिया एक अवधारणा है, यह कोई निश्चित नियम नहीं है।

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Alphabetical Series **Short Notes**

- * $L \quad \underline{L} = -(L) \quad \underline{\text{बायें}} \quad \text{बायें} = -\text{बायें}$
- * $R \quad \underline{R} = -(R) \quad \underline{\text{दायें}} \quad \text{दायें} = -\text{दायें}$
- * $L \quad \underline{R} = +(R) \quad \underline{\text{बायें}} \quad \text{दायें} = +\text{बायें}$
- * $R \quad \underline{L} = +(L) \quad \underline{\text{दायें}} \quad \text{बायें} = +\text{दायें}$

Note: Same आने पर (-) और Different आने पर (+) करते हैं। अगर दो से ज्यादा position हैं तब हिंदी में question आगे से और English में पीछे से solve होगा।

2. Midway (बीच का अक्षर) = Average (औसत)

Note: Average तब निकलता है जब दोनों position same end से हों। अगर ना हो तो बना लो।

Basic Series

1. A B C ... X Y Z

- * $L = 6F$
- * $R = 6(27 - 6 = 21)U$
- * $L = -5(26 - 5 = 21)U$
- * $L = 30(30/26 = 4)D$
- * $R = -5(5 + 1 = 6)F$
- * $R = 30(30/26 = 4; 27 - 4 = 23)W$

2. Z Y X ... C B A

- * $L = 10(27 - 10 = 17)Q$
- * $R = 10J$

3. B D F ... X Z

- * $L = 10 (\times 2 = 20)T$
- * $R = 10 (L = 14 - 10 = 4 \times 2 = 8)H$

4. A C E ... W Y

- * $L = 10 (\times 2 - 1 = 19)S$
- * $R = 10 (L = 14 - 10 = 4(\times 2 - 1) = 7)G$

5. M ... A Z ... N

- * $R = 19(19 - 13 = 6)F$
- * $R = 3(13 + 3 = 16)P$
- * $L = 4(14 - 4 = 10)J$
- * $L = 19(19 - 13 = 6; 27 - 6 = 21)U$

6. M ... A N ... Z

- * $R = 19(19 - 13 = 6)F$
- * $R = 3(27 - 3 = 24)X$
- * $L = 4(14 - 4 = 10)J$
- * $L = 19S$

7. A ... M Z ... N

- * $R = 19(27 - 19 = 8)H$
- * $R = 3(13 + 3 = 16)P$
- * $L = 4D$
- * $L = 19(19 - 13 = 6; 27 - 6 = 21)U$

8. Alphanumeric Series

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Number Series **Short Notes**

Meaning

Series = Numbers arranged in a specific pattern.

Example:

* Whole Numbers

0, 1, 2, 3, 4, 5, ...

* Even Numbers

2, 4, 6, 8, 10, ...

* Odd Numbers

1, 3, 5, 7, 9, ...

* Prime Numbers

2, 3, 5, 7, 11, 13, ...

* Squares

1, 4, 9, 16, 25, 36, ...

* Cubes

1, 8, 27, 64, 125, 216, ...

* Prime - 1

1, 2, 4, 6, 10, 12, 16, ...

Prime + 1

3, 4, 6, 8, 12, 14, 18, ...

} Prime $\pm n$

* Square + 1

2, 5, 10, 17, 26, ...

Cube - 1

0, 7, 26, 63, 124, ...

} Square / Cube $\pm n$

* Multiply

5, 11, 24, 51, 106, ...

1, 1.5, 3, 7.5, ...

} $\times \pm n$

* 3, 10, 31, 80, 171, ...

13, 16, 25, 40, ...

} Steps (Difference)

Important Tricks

* Multiplication

* Difference

* Square / Cube

* Alternate Pattern

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Analogy **Short Notes**

Meaning

Analogy = Relationship between two words/number/letters.

Word Analogy

1. Word Based:

Lion : Forest :: Fish : **Water**

2. Worker & Work Based:

Teacher : School :: Doctor : **Hospital**

3. Part & Whole Based:

Wheel : Car :: Page : **Book**

4. Tool & Work Based:

Pen : Write :: Knife : **Cut**

Numer Analogy

1. 4 : 16 :: 5 : **25** (Square)
2. 7 : 343 :: 9 : **729** (Cube)
3. 8 : 80 :: 6 : **60** (Multiply)
4. 68 : 79 :: 81 : **92** (Addition)
5. 23 : 5 :: 65 : **11** (Digital Sum)

Alphabet Analogy

1. A : C :: B : D
2. ACE : BDF :: MNO : NOP
3. AZ : BY :: CX : DW

Important Tricks

- (i) Relationship पहचानों
- (ii) Same pattern apply करो
- (iii) Word/Number/Alphabet Identify करो

NOTE: Priority

1. Direct Logic
2. Square/Cube
3. Multiplication/Division
4. Addition/Subtraction
5. Digital Sum

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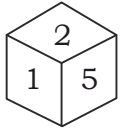
DICE (SHORT NOTES)

1. Standard Dice (मानक पासा):

$1 \leftrightarrow 6, 2 \leftrightarrow 5, 3 \leftrightarrow 4$

1 - 2 - 3
 $\updownarrow \quad \updownarrow \quad \updownarrow$
 5 - 4 - 6

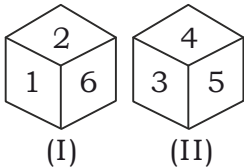
2. One Figure:



$1/2/5 \leftrightarrow 3/4/6$

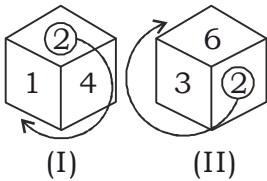
3. Two Figures:

(i) No Common:



$1/2/6 \leftrightarrow 3/4/5$

(ii) One Common:

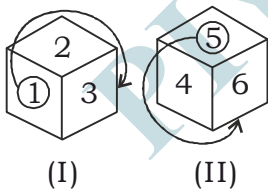


2 - 4 - 1
 5 - 2 - 3 - 6

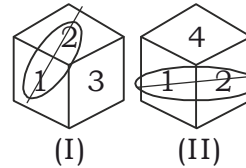
5. Spl. One Common

If 1 is opposite to 5, then:

यदि 1 के विपरीत 5 है तब:



6. Two Common:



$3 \leftrightarrow 4, 1/2 \leftrightarrow 5/6$

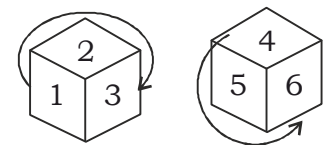
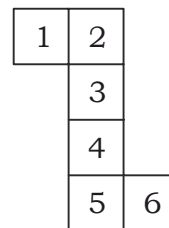
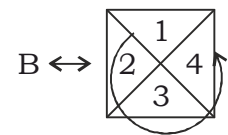
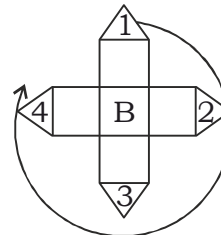
7. Note: Two opposite surfaces can never be seen together and also can never be hidden together.

8. Open Figure:

(i) Alternate:

(ii) ZeeMethod:

9. Rotation Method:



10. More than Two:

कोई भी दो figure से question solve कर सकते हैं।

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Cube And Cuboid (SHORT NOTES)

1. Corner Cubes (शीर्षघन) (3) = 8

Middle/Edge Cubes (मध्यघन) (2) = $12(n - 2)$

12 = Total Edges (किनारे)

Central Cubes (केन्द्रीय घन) (1) = $6(n - 2)^2$

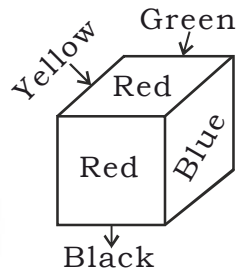
6 = सतह

Inner Central Cubes (अन्तःकेन्द्रीय घन) (0) = $(n - 2)^3$

Total = n^3

$$n = \frac{\text{Big Cube Length}}{\text{Small Cube Length}}$$

2.



$$2[\text{Red} + \text{Blue}] = 2 \times (n - 2)$$

$$2[\text{Red} + \text{Green}] = 1 \times (n - 2)$$

$$1[\text{Red}] = 2 \times (n - 2)^2$$

$$1[\text{Blue}] = 1 \times (n - 2)^2$$

3. **Cube:**

$$3 = 8$$

$$2 = 12(n - 2)$$

$$1 = 6(n - 2)^2$$

$$0 = (n - 2)^3$$

$$T = n^3$$

Cuboid:

$$3 = 8$$

$$2 = 4(n_1 - 2) + 4(n_2 - 2) + 4(n_3 - 2)$$

$$1 = 2(n_1 - 2)(n_2 - 2) + 2(n_2 - 2)(n_3 - 2) + 2(n_3 - 2)(n_1 - 2)$$

$$0 = (n_1 - 2)(n_2 - 2)(n_3 - 2)$$

$$\text{Total} = n_1 \times n_2 \times n_3$$

4. $\frac{2}{2} \times \frac{6}{2} \times \frac{8}{2}$

$$n_1 = 1; n_2 = 3; n_3 = 4$$

$$4 = 4$$

$$3 = T - (4 + 2)$$

$$2 = (n_2 - 2)(n_3 - 2)$$

$$1 = 0$$

$$0 = 0$$

$$T = n_1 \times n_2 \times n_3$$

$$\frac{3}{3} \times \frac{3}{3} \times \frac{15}{3}$$

$$n_1 = 1; n_2 = 1; n_3 = 5$$

$$\text{Total} = n_1 \times n_2 \times n_3$$

$$5 = 2$$

$$4 = \text{Total} - 2$$

$$3 = 0$$

$$2 = 0$$

$$1 = 0$$

$$0 = 0$$

5. $6 \times 8 = \text{Red}$

$$8 \times 10 = \text{Blue}$$

$$10 \times 6 = \text{Yellow}$$

$$\frac{6}{2} \times \frac{8}{2} \times \frac{10}{2}$$

$$n_1 = 3; n_2 = 4; n_3 = 5$$

$$2[\text{Red} + \text{Blue}] = 4(n_2 - 2)$$

$$2[\text{Blue} + \text{Yellow}] = 4(n_3 - 2)$$

$$2[\text{Yellow} + \text{Red}] = 4(n_1 - 2)$$

$$1[\text{Red}] = 2(n_1 - 2)(n_2 - 2)$$

$$1[\text{Blue}] = 2(n_2 - 2)(n_3 - 2)$$

$$1[\text{Yellow}] = 2(n_3 - 2)(n_2 - 2)$$

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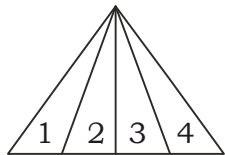
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Counting Figure Short Notes

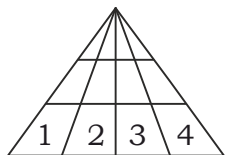
TYPE-1:

1.



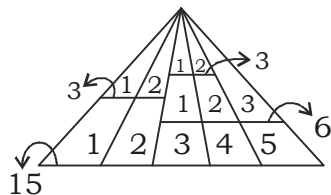
$$\Delta \Rightarrow 1 + 2 + 3 + 4 = 10$$

2.



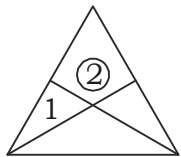
$$\Delta \Rightarrow 10 \times 3 = 30$$

3.



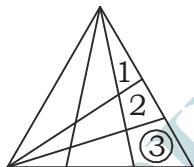
$$\Delta \Rightarrow 15 + 6 + 3 + 3 = 27$$

4.



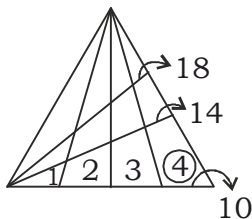
$$\Delta \Rightarrow 2^3 = 8$$

5.



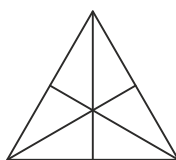
$$\Delta \Rightarrow 3^3 = 27$$

6.



$$\Delta \Rightarrow 18 + 14 + 10 = 42$$

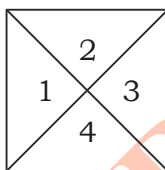
7.



$$\Delta = 16$$

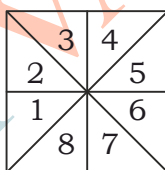
TYPE-2:

8.



$$\Delta \Rightarrow 4 \times 2 = 8$$

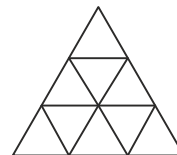
9.



$$\Delta \Rightarrow 8 \times 2 = 16$$

TYPE-3:

10.



$n = \text{even}$

$$\Delta = \frac{n(n+2)(2n+1)}{8}$$

$n = \text{odd}$

$$\Delta = \frac{[n(n+2)(2n+1)-1]}{8}$$

11.

1	2	③
2		
3		
④		

Square = $3 \times 4 + 2 \times 3 + 1 \times 2$

Rectangle:

$$(1 + 2 + 3) \times (1 + 2 + 3 + 4)$$

12. Diagonals

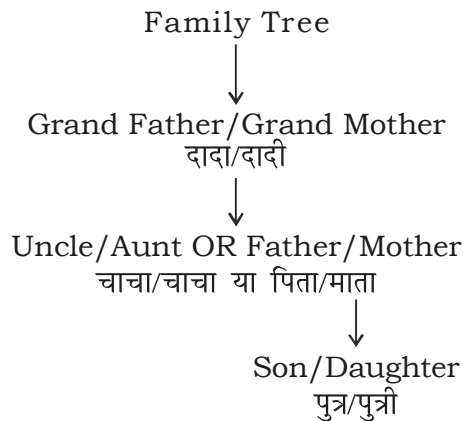
$$D = \frac{n(n-3)}{2}$$

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Blood Relation **Short Notes**



* Puzzle

* Coded

* Pointing

* Generation

* Gender

Male (पुरुष) = Δ

Female (महिला) = $\bigcirc$

English

Uncle

Hindi

चाचा/मामा/अंकल

* Paternal (पिता) Uncle = चाचा/फुफाजी

* Maternal (माता) Uncle = मामा/मौसाजी

* Indirect:

Brother of Father/Father's Brother = पिता का भाई

1. A is the father of B./A, B का पिता है।

$\Delta \rightarrow$ पिता

B $\rightarrow$ पुत्र/पुत्री

2. A is the mother of B./A, B की माँ है।

$\bigcirc$

B

3. A is a son of B./A, B का पुत्र है।

B

Δ

4. A is a daughter of B./A, B की पुत्री है।

B

$\bigcirc$

5. A is a brother of B./A, B का भाई है।

Δ

B $\rightarrow$ बहन/भाई

6. A is a sister of B./B is a brother of A.

A, B की बहन है।/B, A का भाई है।

$\bigcirc$

Δ

7. A is the wife of B./A, B की पत्नी है।

$\bigcirc$

+ Δ

8. A is the paternal uncle (father's brother) of B.

A, B का चाचा है।

Δ

$\Delta \rightarrow$ चाचा

$\Rightarrow$ B

9. A is the grand father of B./A, B का दादा है।

Δ

$\rightarrow$ दादा

Δ

or $\bigcirc$

B

B

10. A is a son-in-law of B./A, B का दामाद है।

B $\rightarrow$ ससुर/सास

$\bigcirc$

+ Δ

11. A is the father-in-law of B./A, B का ससुर है।

Δ

+ Δ

$\bigcirc$

or Δ

+ $\bigcirc$

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Calendar Short Notes

1. Leap Year: $\div 4, \div 400$

2. Years L.Y.

100 $\rightarrow$ 24

200 $\rightarrow$ 48

300 $\rightarrow$ 72

400 $\rightarrow$ 97

3. Odd Days: $\div 7$, (शेषफल)

(i) Days: $\div 7$, Remainder

(ii) Weeks Odd = 0

(iii) 1 N.Y. = 1

1 L.Y. = 2

(iv) Odd Days in Years = Years + L.Y.

4.

Years	L.Y.	Odd Days
100	24	5
200	48	3
300	72	1
400	97	0

5. Age = ?

6. Basic Odd Days Based:

आगे = +

पीछे = -

7. Case I = Difference in Only Years

Case II = Difference in Only Months

Case III = Difference in Years & Months

(*Feb = 28)

8. Repeation of a Normal Year = 11/6

Repeation of a Leap Year = 28

9. **Century First Day**

Monday 100

Saturday 200

Thursday 300

Tuesday 400

Century Last Day

5 Friday

3 Wednesday

1 Monday

0 Sunday

10. **Standard Method**

Total Odd Days

1 = Monday

2 = Tuesday

3 = Wednesday

4 = Thursday

5 = Friday

6 = Saturday

7/0 = Sunday

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Clock Revision Booster

Handwritten Notes

① Image

$$\begin{array}{l} \text{(i) Mirror Image =} \\ 12:00 \\ \underline{12:60} \end{array}$$

$$\begin{array}{l} \text{(ii) Water Image =} \\ 18:30 \\ \underline{17:90} \\ \text{(1 hr less)} \end{array}$$

② Angle

$$\text{Speed of min. hand} = 6^\circ/\text{min}$$

$$\text{Speed of hr. hand} = \frac{1}{2}^\circ/\text{min}$$

$$\text{Speed of Second hand} = 360^\circ/\text{min}$$

$$\text{Angle} = \left(\frac{60H - 11M}{2} \right)$$

$$\text{दूसरा} = 360^\circ - \text{Angle}$$

③ Time

$$\text{mins} = \frac{2}{11} (A_1 \pm A_2)$$

$$A_1 = 30H,$$

$$A_2 = \text{given Angle}$$

⑤ Clock Error

$$\text{(i)} \left(\frac{\frac{5}{6511} - \text{Time}}{\text{Time}} \right) \times \text{T.T.}$$

$$\text{(ii)} \frac{S/F}{S+F} \times \text{T.T.}$$

④ Special Angle

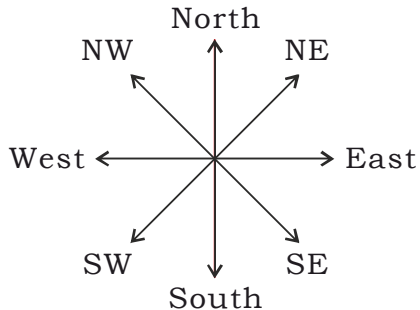
- 0° 12 hrs $\rightarrow$ 12 बर (12)
- 180° 12 hrs $\rightarrow$ 6 बर (6)
- 90° 12 hrs $\rightarrow$ 3, 9 बर (3, 9)

$$\text{(ii)} R \leftrightarrow W$$



Direction And Distance

Short Notes



- 1 Right = 90° CW
1 Left = 90° ACW
2 Left/2 Right = Opposite
3 Right = 1 Left
3 Left = 1 Right
4 Left/4 Right = Same
अगर 4 से ज्यादा Left या Right हो तब 4 से भाग देकर
Remainder निकाल लो।

Type I: North

R
R
L
L
R
? → East

Type II:

? → West
R → Left
L
R
L
R
North

- Type III: (a) East (b) ? → NE
- | | |
|---------|------------------|
| 45° CW | 45° CW |
| 90° CW | 90° ACW |
| 45° ACW | 90° CW |
| 90° ACW | 45° ACW |
| 45° CW | 45° CW = 45° ACW |
| ? → SE | East |

Type IV: W.r.t. Direction & Distance with figure.
Pythagoras Theorem

$$H^2 = P^2 + B^2$$

* 3, 4, 5

* 5, 12, 13

Type V: Direction Change/Rotation

If E $\xrightarrow{45^\circ \text{ CW}}$ S/E

S $\xrightarrow{45^\circ \text{ CW}}$ S/W

Type VI: Head Down & Legs Up

Face = East (उल्टा) or West (सीधा)

R Hand = North

Type VII: Shadow

Sun का L/R = Face = N/S

आगे/पीछे; Face = E/W

Type VIII: Smoke = Opposite to Running Train + Air

Type IX: Coded Direction

Pythagoras Theorem

$$H^2 = P^2 + B^2$$

* 3, 4, 5

* 5, 12, 13

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Syllogism Short Notes

I. All car are Buses. (+)

100 50
सभी कार, बस हैं। (+)

II. Some Rains are Water. (+)

50 50
कुछ वर्षा, पानी है। (+)

III. No Dice is Cube. (-)

100 100
कोई भी पासा, घन नहीं है। (-)

IV. Some Books are not Copies. (-)

50 100
कुछ किताबें, कॉपी नहीं हैं। (-)

V. Only a few A are B. (+, -)

50 50/100
केवल कुछ ही A, B हैं। (+, -)

1. $+$ $\rightarrow$ $+$; $-$ $\rightarrow$ $-$

2. $100 \rightarrow 50$; $50 \nrightarrow 100$

3. $+$ $\rightarrow$ $+$

4. $+$ $\rightarrow$ $-$

5. $-$ $\rightarrow$ No Relation

6. $\oplus$ $\rightarrow$ $\oplus$

7. $\ominus$ $\rightarrow$ $\ominus$

8. Either - or

(a) दोनों निष्कर्ष सही नहीं होने चाहिए।

(b) Sub-Sub और Obj-Obj same होना चाहिए।

(c) एक निष्कर्ष + और दूसरा -ve होना चाहिए।

Either - or (Exceptional/अपवाद)

All + No

Some + Some not

9. Possibility को check करने के लिए हम conclusion बनाते हैं और वह conclusion, decide करता है कि poss. बनेगी या नहीं।

Conclusion $\rightarrow$ Possibility

✓	×
All	×
No	×

(a) Some $\rightarrow$ Some not
All

(b) Some not $\rightarrow$ Some
No

(c) $\times$ ✓

10. Coded Syllogism

11. Reverse/Inverse Syllogism

* All not = Some not (सभी नहीं)

* All can never be = Some not

* All A can never be B.

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Ranking Short Notes

Basic:

$$\text{Total} = \text{Left} + \text{Right} - 1$$

$$\text{Left} = \text{Total} + 1 - \text{Right}$$

$$\text{Right} = \text{Total} + 1 - \text{Left}$$

$$\text{Total} = \text{Top} + \text{Bottom} - 1$$

$$\text{Top} = \text{Total} + 1 - \text{Bottom}$$

$$\text{Bottom} = \text{Total} + 1 - \text{Top}$$

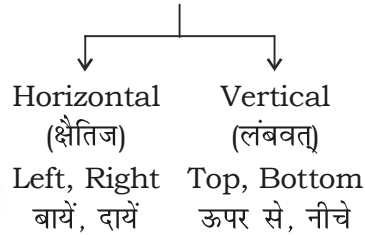
Advance:

$$\text{Rank} = \text{Distance}$$

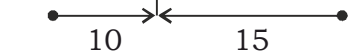
$$\text{Persons} = \text{Distance} - 1$$

[Figure में person से Dis. बनानी है, तो $D = P + 1$]

1. Ranking(क्रम व्यवस्था)

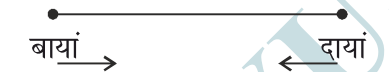


2. L End Swati R End

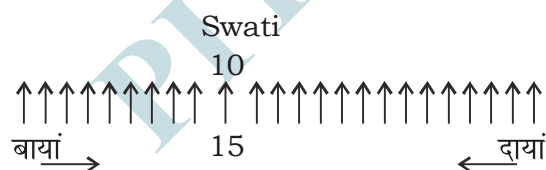


$$\text{Total} = 10 + 15 - 1 = 24$$

L End R End



$$\text{Total} = L + R - 1$$



3. Horizontal

$$\text{Total} = \text{Left} + \text{Right} - 1$$

$$\text{Left} = \text{Total} + 1 - \text{Right}$$

$$\text{Right} = \text{Total} + 1 - \text{Left}$$

$$\text{कुल} = \text{बायें} + \text{दायें} - 1$$

$$\text{बायें} = \text{कुल} + 1 - \text{दायें}$$

$$\text{दायें} = \text{कुल} + 1 - \text{बायें}$$

Vertical

$$\text{Total} = \text{Top} + \text{Bottom} - 1$$

$$\text{Top} = \text{Total} + 1 - \text{Bottom}$$

$$\text{Bottom} = \text{Total} + 1 - \text{Top}$$

$$\text{कुल} = \text{ऊपर} + \text{नीचे} - 1$$

$$\text{ऊपर} = \text{कुल} + 1 - \text{नीचे}$$

$$\text{नीचे} = \text{कुल} + 1 - \text{ऊपर}$$

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Seating Arrangement

Short Notes

In Seating arrangement questions we are given some conditions and have to arrange a group of people according to those conditions. There are mainly two types of arrangements that are asked in the examination.

इसमें आने वाले प्रश्नों में कुछ शर्तें दी जाती हैं और उन शर्तों को पूरी करते हुए हमें व्यक्तियों के समूह को व्यवस्थित करना होता है। व्यक्तियों के समूह के बैठने की व्यवस्था मुख्यतः दो प्रकार की होती है-

1. Close form/बंद व्यवस्था
2. Open form/खुली व्यवस्था

First we will learn to solve the close form questions. सबसे पहले हम बंद व्यवस्था से सम्बन्धित प्रश्नों को हल करना सीखेंगे।

1. Close Form/बंद व्यवस्था:

Such an arrangement is closed from all sides. For eg.-a group of people can sit around a circle. Similarly square, rectangle, pentagon etc. are also included in close form arrangements.

ऐसी व्यवस्था चारों ओर से बंद होती है। जैसे-हम व्यक्तियों के समूह को वृत्त के चारों ओर बैठा सकते हैं। इसी प्रकार वर्गाकार, आयताकार, पंचभुजाकार आदि भी बंद व्यवस्था में ही आएंगे।

To solve close form questions we should know the meaning of some basic words.

बंद व्यवस्था से संबंधित प्रश्नों को हल करने के लिए सबसे पहले हम कुछ आधारभूत शब्दों का अर्थ जानेंगे।

(1) Left-right/बायाँ-दायाँ

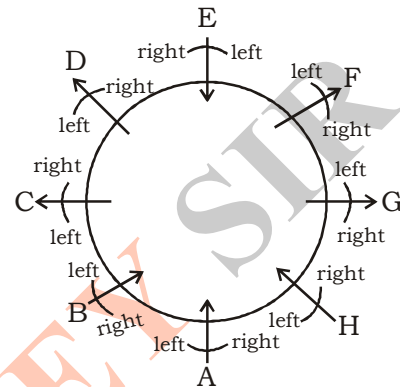
The left-right of a person depend on the direction of his face.

किसी व्यक्ति का बायाँ-दायाँ उस व्यक्ति के मुख की दिशा पर निर्भर करता है।

For e.g. 8 persons A, B, C, D, E, F, G and H are sitting around a circle. A, B, E and H are facing the centre while C, D, F and G are facing opposite to the centre.

जैसे- आठ व्यक्ति A, B, C, D, E, F, G और H एक वृत्त के चारों ओर बैठे हैं। इनमें से A, B, E और H केंद्र की ओर मुँह करके बैठे हैं जबकि C, D, F और G केंद्र से बाहर की ओर मुँह करके

बैठे हैं।



To the right of A = H ; To the left of A = B
A के दाएँ है : H ; A के बाएँ है : B
To the right of C = D ; To the left of C = B
C के दाएँ है : D ; C के बाएँ है : B

(2) Immediate Neighbour/पड़ोसी:

The immediate neighbours of a person are those who are sitting close to that person. For e.g. in the above figure-

किसी व्यक्ति के पड़ोसी वह व्यक्ति होंगे जो उस व्यक्ति के बिल्कुल पास बैठे हैं। जैसे उपर्युक्त चित्र में-

The immediate neighbours of A = B & H
A के पड़ोसी = B व H

The immediate neighbours of G = F & H
G के पड़ोसी = F व H

The immediate neighbours of D = C & E
D के पड़ोसी = C व E

Similarly we will find out the immediate neighbours of other persons.

इसी प्रकार अन्य व्यक्तियों के पड़ोसी ज्ञात करेंगे।

(3) Next to/अगले स्थान पर:

The words "Next to" and "Immediate Neighbour" have the same meaning. For e.g. in the above figure- "अगले स्थान पर" और "पड़ोसी" दोनों का अर्थ बिल्कुल एक जैसा होता है। जैसे उपर्युक्त चित्र में-

Next to A are : B & H ; Next to C are : B&D ; Next to F are : E & G

A के अगले स्थान पर: B व H ; C के अगले स्थान पर : B व D
F के अगले स्थान पर : E व G

Similarly we can find out it for other persons.

इसी प्रकार अन्य व्यक्तियों के अगले स्थान पर बैठे व्यक्ति ज्ञात कर सकते हैं।

(4) Between/बीच में:

If three persons are sitting together then the person sitting in the middle position is said to be the between of the other two persons. For e.g. in the above figure-

किन्हीं तीन एक साथ बैठे व्यक्तियों में मध्य स्थान पर बैठा व्यक्ति अन्य दो व्यक्तियों के ठीक बीच में कहलाता है। जैसे उपर्युक्त चित्र में-

Between B and H : A ; Between C and E : D

B व H के बीच में : A ; C व E के बीच में : D

Between A and G : H

A व G के बीच में : H

Similarly we can find out it for other persons.

इसी प्रकार अन्य के लिए भी बीच में व्यक्ति ज्ञात कर सकते हैं।

(5) Opposite/विपरीत:

If two persons are sitting in the opposite directions, they are called opposite to each other. Such people may or may not be facing each other. For e.g. in the above figure.

दो व्यक्ति जो एक-दूसरे की विपरीत दिशाओं में बैठे हैं अर्थात् आमने-सामने बैठे हैं। ऐसे व्यक्ति एक-दूसरे के सम्मुख हो भी सकते हैं और नहीं भी। जैसे उपर्युक्त चित्र में-

A and E are opposite to each other.

A और E एक-दूसरे के विपरीत हैं।

B and F are opposite to each other.

B और F एक-दूसरे के विपरीत हैं।

C and G are opposite to each other.

C और G एक-दूसरे के विपरीत हैं।

D and H are opposite to each other.

D और H एक-दूसरे के विपरीत हैं।

(6) Facing/सम्मुख:

Facing means two persons are sitting in the opposite directions and facing each other. For e.g. in the

above figure-

दो व्यक्ति जो एक-दूसरे की विपरीत दिशाओं में बैठे हैं और एक-दूसरे की तरफ मुख करके बैठे हैं। जैसे उपर्युक्त चित्र में-
A and E are facing each other.

A और E एक-दूसरे के सम्मुख हैं।

Note/नोट:

B&F, C&G and D&H are opposite to each other but they are not facing each other.

यहाँ B व F, C व G और D व H एक-दूसरे के सम्मुख नहीं बैठे हैं। वे केवल एक-दूसरे के विपरीत बैठे हैं।

(7) Some other words/कुछ अन्य शब्द:-

(i) Who, which, that-These words are used for object.
'जो' (who, which, that)-इन शब्दों का प्रयोग कर्म (object) के लिए होता है।

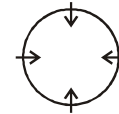
(ii) And, while, whereas, But-These words are used for subject.

'और' (and), 'जबकि' (while, whereas), 'लेकिन' (But)-इन शब्दों का प्रयोग कर्ता (Subject) के लिए होता है।

There are mainly three types of close form arrangements:

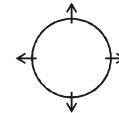
बन्द व्यवस्था से संबंधित व्यवस्थाएँ मुख्यतः तीन प्रकार की होती हैं-

(1) When all the people are sitting facing the centre-
जब सभी व्यक्ति केंद्र की ओर मुँह करके बैठे हो-



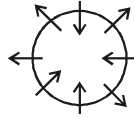
(2) When all the people are sitting facing opposite to the centre-

जब सभी व्यक्ति केंद्र से बाहर की ओर मुँह करके बैठे हो-



(3) Mixed Arrangements: When some persons are facing the centre while others are facing opposite to the centre, it is called mixed arrangement.

मिश्रित व्यवस्था: जब कुछ व्यक्ति केंद्र की ओर मुँह करके बैठे हैं और कुछ व्यक्ति केंद्र से बाहर की ओर मुँह करके बैठे हैं तो इसे मिश्रित व्यवस्था कहते हैं।



Note/नोट:

When it is not mentioned in the question whether all the persons are facing centre or opposite to the centre, we will solve the question by assuming the face of all the people towards the centre.

जब प्रश्न में यह नहीं बताया जाए कि व्यक्तियों का मुँह केंद्र की ओर है या केंद्र से बाहर की ओर है तो हम सभी व्यक्तियों का मुँह केंद्र की ओर मानकर प्रश्न हल करेंगे।

To solve the close form questions we follow the steps given below:

बंद व्यवस्था से संबंधित प्रश्नों को निम्न तरह से हल करेंगे-

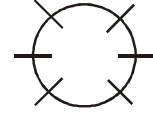
- (1) First read the given data carefully. After it draw the shape mentioned in the question. For e.g.-If all people are sitting around the circle, we will draw a circle. Similarly if all people are sitting around a rectangular table, we will draw rectangle.

सबसे पहले प्रश्न में दी गई सूचनाओं को ध्यान से पढ़ें। उसके बाद वह आकृति बनाएं जिसका प्रश्न में उल्लेख किया गया है। जैसे- यदि सभी व्यक्ति वृत्त के चारों ओर बैठे हैं तो वृत्त बनाएंगे। यदि सभी व्यक्ति आयत के चारों ओर बैठे हैं तो आयत बनाएंगे।

- (2) Now, Considering all the people in the question, mark the slots as given below:
अब प्रश्न में जितने व्यक्ति हैं उन सभी व्यक्तियों को एक-साथ बैठा दें।

Here 6 persons are sitting around the circle.

जैसे-छ: व्यक्ति एक वृत्त के चारों ओर बैठे हैं तो उसे निम्न प्रकार दर्शाएंगे-



- (3) If it is known whether people are facing the centre or opposite to the centre, then we will also mark it. यदि सभी व्यक्तियों के मुँह की दिशा ज्ञात हो तो आकृति में वह भी चिह्नित कर दें।

- (4) In close form question we can make the first person sit anywhere as per our wish. After it using all definite points, we will fill up as many empty places as possible.

बंद व्यवस्था के प्रश्नों में हम पहले व्यक्ति को अपनी इच्छानुसार कहीं भी बैठा सकते हैं और उसके बाद प्रश्न में दिए गए निश्चित बिन्दुओं को प्रयोग करते हुए ज्यादा से ज्यादा खाली स्थानों पर अन्य व्यक्तियों को बैठाएंगे।

- (5) If there are two or more possibilities of seating individuals while solving the question, then we have to make all such possibilities. In the end the possibility which does not violate any condition, will be correct.

यदि प्रश्न को हल करते समय व्यक्तियों को बैठाने की दो या अधिक संभावनाएँ बनें तो हमें ऐसी सभी संभावनाएँ बनानी होंगी। अंत में वह संभावना सही होगी जो प्रश्न की शर्तों के विरुद्ध नहीं हो।

Piyush Varshney Sir (Reasoning Expert)

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